

Quenched Pair-Coupling Disorder in Frustration-Free QGN Parents II

Quenched dynamical-source fluctuations, sharp cutoffs, and disorder-noise geometry

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Abstract

This companion paper studies quenched fluctuations of the low-energy dynamical source tail for the periodic random-conductance operator arising from the weighted QGN parent of Paper I. For product i.i.d. conductances on Λ_L , the physical sharp observable is

$$T_L(\delta) = L^{-d} \left\langle g_L, L_{J,L}^+ \mathbf{1}_{(0,\delta]}(L_{J,L}) g_L \right\rangle.$$

The spectral projector is discontinuous in each bond variable. We therefore introduce the positive Abel envelope $\mathfrak{A}_L(\delta) = L^{-d} \langle g_L, L_{J,L}^+ e^{-L_{J,L}/\delta} g_L \rangle$ and derive an exact bond-sensitivity identity. Under a one-bond Poincaré inequality, a period-uniform gradient-propagator estimate gives

$$\text{Var} \mathfrak{A}_L(\delta) \leq C j_+^2 |A|^4 L^{-d} (\delta/j_+)^{d/2}.$$

This yields a one-sided quenched bound for T_L . The same second moment gives the target variance scale at every bounded effective-mode cutoff only through $\text{Var} T_L \leq \mathbb{E} T_L^2$; it is not a mesoscopic concentration theorem. In the frozen-spectrum Born sector, the sharp tail is an exact quadratic form and obeys Hanson–Wright concentration.

We audit three routes to a nonlinear fixed-cutoff theorem. A finite discrete relative Wegner estimate controls eigenvalue counts but a modulated cycle shows that counts do not control current weight. Exponential threshold randomization is exactly the Abel envelope, while positive heat-mixture thresholds cannot be relatively narrow. Logarithmic shell averaging gives the optimal shell scale and a deterministic interval argument removes exceptional cutoffs only at the uncentered floor. Rank-one interlacing proves that one bond crosses a prescribed cutoff at most once; Krein coordinates and a coarea identity determine the exact current-weighted crossing surface. An acoustic surrogate shows why a vanishing uniform maximal-shell modulus is too strong. The remaining mesoscopic theorem is isolated as a fixed-cutoff disorder-noise quadratic-variation bound with an integrable $t^{-1/2}$ boundary layer.

Claim hierarchy.

1. The exact QGN-to-random-conductance reduction and annealed source-tail estimate are stated as inputs from Paper I.
2. The Abel sensitivity identity, gradient-propagator reduction, smooth-tail variance, and one-sided sharp-tail theorem are proved here.
3. The bounded-mode sharp variance uses only $\text{Var} T \leq \mathbb{E} T^2$ and is not claimed as mesoscopic centering.

4. The Born concentration, relative Wegner estimate, randomization identities, log-shell theorem, one-crossing geometry, Krein formulas, participation coarea, and noise-semigroup identities are proved here.
5. The remaining open input is the fixed-cutoff disorder-noise estimate stated in Problem 7.1; no forward equation reference is used in this front matter.

1 Setting and inputs from Paper I

Let $d \geq 2$, $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^d$, $N = L^d$, and orient each nearest-neighbor bond in the positive coordinate direction. The conductances on the dL^d torus bonds are independent with common law $\beta \subset [j_-, j_+]$ and are extended periodically. This is the product-torus ensemble, not a coupled periodization of one infinite realization. One-dimensional rings appear below only as exactly solvable obstructions.

For $A \in \mathbb{R}^d$, define $a_{x,\mu} = A_\mu$,

$$g_L(x) = \sum_{\mu=1}^d (j_{x-e_\mu,\mu} - j_{x,\mu}) A_\mu, \quad (1)$$

$$(L_{J,L}f)(x) = \sum_{\mu=1}^d [j_{x,\mu}(f(x) - f(x + e_\mu)) + j_{x-e_\mu,\mu}(f(x) - f(x - e_\mu))]. \quad (2)$$

Thus $g_L = BJ_LRA$ and $L_{J,L} = BJ_LB^*$. Let τ_x translate the periodic environment and define the horizontal differences

$$D_\mu F(J) = F(\tau_{e_\mu} J) - F(J), \quad D_\mu^* F(J) = F(\tau_{-e_\mu} J) - F(J). \quad (3)$$

The environment generator and local drift are

$$\mathcal{L}_L = D^* j(0) D, \quad d_A = D^*(j(0)A). \quad (4)$$

The stationary extension of d_A is g_L . Paper I proves the exact site/target identity and the total-energy relation

$$\langle g_L, L_{J,L}^+ g_L \rangle = \sum_e j_e a_e^2 - \mathcal{E}_{J_L}[a]. \quad (5)$$

It also proves the deterministic gap

$$\lambda_{\text{gap},L} = 4j_- \sin^2(\pi/L) \quad (6)$$

and the period-uniform annealed estimates

$$\mathbb{E} N^{-1} \langle g_L, e^{-tL_{J,L}} g_L \rangle \leq C j_+^2 |A|^2 (1 + j_+ t)^{-1-d/2}, \quad (7)$$

$$\mathbb{E} T_L(\delta) \leq C j_+ |A|^2 \min\{1, (\delta/j_+)^{d/2}\}. \quad (8)$$

We restate these inputs to make the fluctuation arguments self-contained at the operator level; the many-body derivation is in Paper I [1].

Theorem 1.1 (Imported exact and annealed input from Paper I). *Equations (5)–(8) hold uniformly in L . The exact environment–site intertwining identifies the stationary extension of $e^{-t\mathcal{L}_L} d_A$ with $e^{-tL_{J,L}} g_L$.*

2 Abel smoothing, variance, and the sharp edge

The annealed theorem controls the mean of the physical sharp tail

$$T_L(\delta) := \frac{1}{L^d} \left\langle g_L, L_{J,L}^+ \mathbf{1}_{(0,\delta]}(L_{J,L}) g_L \right\rangle. \quad (9)$$

A centered fluctuation proof cannot begin by differentiating this expression: the map $J \mapsto \mathbf{1}_{(0,\delta]}(L_{J,L})$ jumps when an eigenvalue crosses the cutoff. We first replace the projector by a positive smooth envelope, prove the optimal effective-mode variance scale for that envelope, and then return to the sharp observable by a samplewise comparison.

Define the Abel tail

$$\mathfrak{A}_L(\delta) := \frac{1}{L^d} \left\langle g_L, L_{J,L}^+ e^{-L_{J,L}/\delta} g_L \right\rangle \quad (10)$$

$$= \frac{1}{L^d} \int_{1/\delta}^{\infty} \left\langle g_L, e^{-tL_{J,L}} g_L \right\rangle dt. \quad (11)$$

Since $e^{-\lambda/\delta} \geq e^{-1}$ for $0 < \lambda \leq \delta$, one has the realization-wise envelope

$$\boxed{T_L(\delta) \leq e \mathfrak{A}_L(\delta)}. \quad (12)$$

Integrating (7) also gives

$$\mathbb{E} \mathfrak{A}_L(\delta) \leq C j_+ |A|^2 \left(\frac{\delta}{j_+} \right)^{d/2}, \quad 0 < \delta \leq j_+. \quad (13)$$

The envelope has an exact cell-problem interpretation. Put

$$\mathcal{E}_J(\phi) = \frac{1}{L^d} \sum_e j_e (a_e + (B^* \phi)_e)^2, \quad \phi_* = -L_{J,L}^+ g_L, \quad (14)$$

and define the parabolic corrector

$$\phi_t = -(\mathbb{I} - e^{-tL_{J,L}}) L_{J,L}^+ g_L. \quad (15)$$

Lemma 2.1 (Parabolic energy-excess identity). *For every realization and $t \geq 0$,*

$$\boxed{\mathcal{E}_J(\phi_t) - \mathcal{E}_J(\phi_*) = \frac{1}{L^d} \left\langle g_L, L_{J,L}^+ e^{-2tL_{J,L}} g_L \right\rangle}. \quad (16)$$

Hence $\mathfrak{A}_L(\delta)$ is exactly the residual cell-energy error after parabolic relaxation for time $(2\delta)^{-1}$.

Proof. Completing the square around the minimizer gives

$$\mathcal{E}_J(\phi) - \mathcal{E}_J(\phi_*) = \frac{1}{L^d} \langle \phi - \phi_*, L_{J,L}(\phi - \phi_*) \rangle.$$

Here $\phi_t - \phi_* = e^{-tL_{J,L}} L_{J,L}^+ g_L$. The spectral theorem yields (16). $\square$

A rational positive envelope will also be useful. Define

$$\mathfrak{R}_{L,1}(\delta) = \frac{1}{L^d} \left\langle g_L, \frac{\delta^2}{L_{J,L}(L_{J,L} + \delta)^2} g_L \right\rangle. \quad (17)$$

Because $\delta^2/(\lambda + \delta)^2 \geq 1/4$ on $0 < \lambda \leq \delta$,

$$\boxed{T_L(\delta) \leq 4 \mathfrak{R}_{L,1}(\delta)}. \quad (18)$$

This is the first positive modified-corrector/Richardson spectral bias of Gloria and Mourrat [2].

2.1 Exact vertical derivative and the gradient propagator

Let $b_e = B e_e$ be the incidence column of edge e and set

$$w_e(t) = b_e^* e^{-tL_{J,L}} g_L, \quad \tau = \delta^{-1}. \quad (19)$$

Lemma 2.2 (Exact Abel-tail sensitivity). *At every coefficient configuration in the interior of the ellipticity box,*

$$L^d \partial_{j_e} \mathfrak{A}_L(\delta) = 2a_e \int_{\tau}^{\infty} w_e(s) ds - \int_{\tau}^{\infty} \int_0^s w_e(s-r) w_e(r) dr ds. \quad (20)$$

Proof. Use (11), $\partial_{j_e} g_L = a_e b_e$, and $\partial_{j_e} L_{J,L} = b_e b_e^*$. Duhamel's formula gives

$$\partial_{j_e} e^{-sL_{J,L}} = - \int_0^s e^{-(s-r)L_{J,L}} b_e b_e^* e^{-rL_{J,L}} dr.$$

Self-adjointness turns the two matrix elements in the second term into $w_e(s-r)w_e(r)$. $\square$

The new probabilistic input is a gradient upgrade of the divergence-semigroup bound already used. Related environment-process decay estimates are discussed in [5]; the precise period-uniform statements and the weighted-Hölder argument needed here are collected in Appendix A.

Lemma 2.3 (Periodic gradient-propagator estimate). *Under the periodic i.i.d. uniformly elliptic hypotheses, for $p = 1, 2$,*

$$\left(\mathbb{E} \frac{1}{dL^d} \sum_e |w_e(t)|^{2p} \right)^{1/(2p)} \leq C_p j_+ |A| (1 + j_+ t)^{-1-d/4}. \quad (21)$$

The same conclusion holds for a periodic mixing ensemble whenever the high-moment divergence-semigroup and weighted parabolic Green-gradient estimates used below are uniform in the period.

Proof. Appendix A defines the torus Green function and weight, restates the two imported GNO estimates with their exact hypotheses and source numbers, and gives the weighted-Hölder calculation in full. The constants are uniform in L . $\square$

2.2 CLT-order variance of the smooth tail

For the centered theorem we impose one additional, explicit regularity assumption on the one-edge law β . Suppose

$$\text{Var}_{\beta} f \leq C_{\beta} \int |f'(s)|^2 d\beta(s), \quad C_{\beta} \leq C_0 j_+^2. \quad (22)$$

The product law then tensorizes. Uniform and smooth positive densities on a fixed ellipticity interval satisfy such an inequality. Discrete laws require an oscillation/Efron–Stein version and are not included in the derivative statement below.

Theorem 2.4 (Quenched Abel-tail variance). *Assume the hypotheses of theorem 2.3 and the one-edge Poincaré inequality (22). For $0 < \delta \leq j_+$,*

$$\text{Var} \mathfrak{A}_L(\delta) \leq C j_+^2 |A|^4 L^{-d} \left(\frac{\delta}{j_+} \right)^{d/2}. \quad (23)$$

The constant depends only on d , j_-/j_+ , and C_0 .

Proof. Normalize $j_+ = 1$ and put $h(t) = (1+t)^{-1-d/4}$. Minkowski's inequality and (21) with $p = 1$ give

$$\left(\mathbb{E} \frac{1}{dL^d} \sum_e \left| \int_\tau^\infty w_e(s) ds \right|^2 \right)^{1/2} \leq C |A| \int_\tau^\infty h(s) ds \leq C |A| \tau^{-d/4}. \quad (24)$$

For the quadratic term, use $p = 2$ and Hölder in probability:

$$\|w_e(r)w_e(u)\|_2 \leq \|w_e(r)\|_4 \|w_e(u)\|_4.$$

Moreover,

$$\int_{r,u \geq 0: r+u \geq \tau} h(r)h(u) dr du \leq 2 \|h\|_1 \int_{\tau/2}^\infty h(s) ds \leq C \tau^{-d/4}. \quad (25)$$

Applying Minkowski once more therefore bounds the averaged $L^2(\Omega_L)$ norm of the double integral in (20) by $C |A|^2 \tau^{-d/4}$. Since $|a_e| \leq |A|$,

$$\frac{1}{dL^d} \sum_e \mathbb{E} |\partial_{j_e} \mathfrak{A}_L(\delta)|^2 \leq C L^{-2d} |A|^4 \tau^{-d/2}. \quad (26)$$

Tensorized Poincaré and summation over the dL^d bonds give $\text{Var} \mathfrak{A}_L \leq C |A|^4 L^{-d} \delta^{d/2}$. Finally, $L_{J,L} = j_+ L \tilde{j}_{J,L}$ and $g_L = j_+ g \tilde{j}_{J,L}$ imply

$$\mathfrak{A}_L^J(\delta) = j_+ \mathfrak{A}_L^{\tilde{J}}(\delta/j_+),$$

which restores (23). $\square$

Remark 2.5 (Higher moments). Combining the same sensitivity estimate with a tensorized high-moment Poincaré inequality gives, for every fixed $p < \infty$,

$$\|\mathfrak{A}_L - \mathbb{E} \mathfrak{A}_L\|_{L^{2p}} \leq C_p j_+ |A|^2 L^{-d/2} \left(\frac{\delta}{j_+} \right)^{d/4}. \quad (27)$$

We do not claim an exponential tail without separately auditing the growth of C_p .

Corollary 2.6 (Quenched one-sided sharp-tail bound). *Under the hypotheses of theorem 2.4, for every $0 < \vartheta < 1$, with probability at least $1 - \vartheta$,*

$$T_L(\delta) \leq C j_+ |A|^2 \left[\left(\frac{\delta}{j_+} \right)^{d/2} + \vartheta^{-1/2} L^{-d/2} \left(\frac{\delta}{j_+} \right)^{d/4} \right]. \quad (28)$$

In addition,

$$\mathbb{E} T_L(\delta)^2 \leq C j_+^2 |A|^4 \left[\left(\frac{\delta}{j_+} \right)^d + L^{-d} \left(\frac{\delta}{j_+} \right)^{d/2} \right]. \quad (29)$$

Proof. Chebyshev applied to theorem 2.4, together with (13) and the samplewise envelope (12), proves (28). The same envelope and $\mathbb{E} \mathfrak{A}_L^2 = (\mathbb{E} \mathfrak{A}_L)^2 + \text{Var} \mathfrak{A}_L$ give (29). $\square$

Define the effective number of diffusive soft modes

$$m_{\text{eff}}(L, \delta) = L^d \left(\frac{\delta}{j_+} \right)^{d/2}. \quad (30)$$

The ratio of the fluctuation term in (28) to the natural annealed scale $(\delta/j_+)^{d/2}$ is $m_{\text{eff}}^{-1/2}$. Thus the fluctuation correction is lower order for $\delta/j_+ \gg L^{-2}$; whenever the mean has a nondegenerate $\delta^{d/2}$ coefficient, this is relative self-averaging. At $\delta \asymp j_+ L^{-2}$ only order-one effective modes are sampled and order-one relative fluctuations are natural. Below (6), the sharp tail is exactly zero.

Corollary 2.7 (Every-cutoff centered bound in the bounded-mode window). *Under the hypotheses of theorem 2.4, define*

$$V_L(\delta) = j_+^2 |A|^4 L^{-d} \left(\frac{\delta}{j_+} \right)^{d/2}. \quad (31)$$

Then, at every deterministic cutoff,

$$\boxed{\text{Var } T_L(\delta) \leq C(1 + m_{\text{eff}}(L, \delta)) V_L(\delta).} \quad (32)$$

Consequently, for each fixed $M < \infty$,

$$m_{\text{eff}}(L, \delta) \leq M \implies \text{Var } T_L(\delta) \leq C_M V_L(\delta). \quad (33)$$

No cancellation around the mean is used here: the estimate is the elementary inequality $\text{Var } T_L \leq \mathbb{E} T_L^2$. It has the centered target scale only when m_{eff} is bounded and becomes noninformative for the mesoscopic many-mode regime. In particular, the desired centered nonlinear sharp-cutoff scale already holds at every cutoff in any fixed acoustic window $\delta \leq C_0 j_+ L^{-2}$; below the deterministic gap the observable vanishes exactly.

Proof. Since variance is bounded by the second moment, rewrite (29) using $m_{\text{eff}} = L^d(\delta/j_+)^{d/2}$:

$$j_+^2 |A|^4 (\delta/j_+)^d = m_{\text{eff}}(L, \delta) V_L(\delta).$$

This proves (32) and its consequences. $\square$

This observation removes the bottom of the diffusive window from the open problem. Genuine pointwise centering is needed only in the mesoscopic many-mode regime $m_{\text{eff}} \rightarrow \infty$.

3 Exact centered theorem in the Born sector

The sharp projector can be treated without smoothing in the frozen-spectrum Born sector. Put

$$J_\varepsilon = \mathbb{I} + \varepsilon \text{diag}(\xi_e), \quad \mathbb{E} \xi_e = 0, \quad (34)$$

and let $L_0 = BB^*$ be the clean torus Laplacian. Since $Ba = 0$, the exact first-order drift is $g_\varepsilon = \varepsilon B \text{diag}(\xi) a$. Define

$$T_L^B(\delta) = \frac{\varepsilon^2}{L^d} \xi^T M_\delta \xi, \quad M_\delta = D_a B^* L_0^+ \mathbf{1}_{(0, \delta]}(L_0) B D_a, \quad (35)$$

where $D_a = \text{diag}(a_e)$. If δ is not a positive eigenvalue of L_0 , then $T_L^B(\delta)$ is the ε^2 coefficient of the full sharp tail as $\varepsilon \rightarrow 0$. At a clean eigenvalue threshold the discontinuous sharp functional need not possess a two-sided Taylor coefficient; (35) is then the canonical frozen-spectrum Born observable, and the theorem below concerns that observable exactly.

Theorem 3.1 (Born sharp-tail concentration). *Let the ξ_e be independent, centered, and sub-Gaussian with $\|\xi_e\|_{\psi_2} \leq K$. For $0 < \delta \leq 1$,*

$$\|M_\delta\| \leq |A|^2, \quad \|M_\delta\|_{\mathbb{F}}^2 \leq C_d |A|^4 L^d \delta^{d/2}, \quad (36)$$

and

$$\boxed{\mathbb{P} \left[\left| T_L^B(\delta) - \mathbb{E} T_L^B(\delta) \right| \geq r \right] \leq 2 \exp \left[-c \min \left\{ \frac{L^d r^2}{\varepsilon^4 K^4 |A|^4 \delta^{d/2}}, \frac{L^d r}{\varepsilon^2 K^2 |A|^2} \right\} \right].} \quad (37)$$

In particular,

$$\text{Var } T_L^B(\delta) \leq C\varepsilon^4 K^4 |A|^4 L^{-d} \delta^{d/2}. \quad (38)$$

If $\sigma^2 = \mathbb{E}\xi_e^2$, then the exact mean is

$$\mathbb{E}T_L^B(\delta) = \frac{\varepsilon^2 \sigma^2}{L^d} \sum_{\substack{q \neq 0 \\ \lambda(q) \leq \delta}} \frac{\sum_{\mu=1}^d A_\mu^2 4 \sin^2(q_\mu/2)}{\sum_{\nu=1}^d 4 \sin^2(q_\nu/2)}. \quad (39)$$

Proof. The operator inequality $BD_a^2 B^* \preceq |A|^2 BB^*$ gives the norm bound in (36). The rank of M_δ is at most the number of clean torus eigenvalues in $(0, \delta]$, which is bounded by $C_d L^d \delta^{d/2}$ for $0 < \delta \leq 1$. Hence $\|M_\delta\|_F^2 \leq \text{rank}(M_\delta) \|M_\delta\|^2$. The Hanson–Wright inequality [6] applied to (35) proves (37) and (38). Finally, Fourier diagonalization of L_0 and $\mathbb{E}\xi_e \xi_{e'} = \sigma^2 \delta_{ee'}$ gives (39). $\square$

Remark 3.2 (Exact Born variance formula). If $\kappa_4 = \mathbb{E}\xi_e^4 - 3\sigma^4$, then

$$\text{Var } T_L^B = \frac{\varepsilon^4}{L^{2d}} \left[2\sigma^4 \text{tr}(M_\delta^2) + \kappa_4 \sum_e (M_\delta)_{ee}^2 \right]. \quad (40)$$

This is the formula checked by the numerical audit.

4 Three closure routes: exact progress and sharp obstructions

The smooth variance theorem leaves a precise question: can one transfer the centered estimate to the discontinuous cumulative function (9)? We now audit the three proposed routes. Two of them yield exact new representations, and the first yields a finite-volume Wegner theorem, but the analysis also shows why none of the standard count-level inputs alone closes the fixed-cutoff theorem.

For a Borel set $I \subset (0, \infty)$, write

$$\mu_{L,J}(I) = \frac{1}{L^d} \left\langle g_L, L_{J,L}^+ \mathbf{1}_I(L_{J,L}) g_L \right\rangle. \quad (41)$$

Thus $T_L(\delta) = \mu_{L,J}((0, \delta])$ and $S_L(\delta, \eta) = \mu_{L,J}((\delta, (1 + \eta)\delta])$.

Route I: a relative Wegner estimate. Assume in this paragraph that the edge variables are independent with common density ρ supported in $[j_-, j_+]$, and put $\rho_\infty = \|\rho\|_\infty$. Let

$$\mathcal{N}_L(I) = \text{Tr}_{\mathbf{1}^\perp} \mathbf{1}_I(L_{J,L}) \quad (42)$$

be the positive-eigenvalue count.

Theorem 4.1 (Finite discrete relative Wegner estimate). *For every interval $I = (E, E + \ell] \subset (0, \infty)$,*

$$\mathbb{E}\mathcal{N}_L(I) \leq dL^d \rho_\infty j_+ \frac{\ell}{E}. \quad (43)$$

Together with ellipticity and the clean-torus Weyl count,

$$\mathbb{E}\mathcal{N}_L(I) \leq C_d L^d \min \left\{ \left(\frac{E + \ell}{j_-} \right)^{d/2}, \rho_\infty j_+ \frac{\ell}{E} \right\}. \quad (44)$$

Proof. Put $L_0 = BB^* = \sum_e b_e b_e^*$ and choose the nondecreasing ramp

$$F(\lambda) = \begin{cases} 0, & \lambda \leq E, \\ \lambda - E, & E < \lambda < E + \ell, \\ \ell, & \lambda \geq E + \ell. \end{cases}$$

Mollify the piecewise-linear ramp F in arbitrarily small neighborhoods of its two corners, apply the preceding identity to the smooth monotone approximants, and pass to the limit by monotone convergence on both sides. Since $L_{J,L} \preceq j_+ L_0$,

$$\mathcal{N}_L(I) \leq E^{-1} \operatorname{Tr}[L_{J,L} \mathbf{1}_I(L_{J,L})] \leq \frac{j_+}{E} \operatorname{Tr}[L_0 \mathbf{1}_I(L_{J,L})].$$

Functional differentiation gives

$$\sum_e \partial_{j_e} \operatorname{Tr} F(L_{J,L}) = \operatorname{Tr}[F'(L_{J,L}) L_0] = \operatorname{Tr}[\mathbf{1}_I(L_{J,L}) L_0].$$

Condition on every coefficient except $j_e = s$. Since $\partial_s L_{J(e,s),L} = b_e b_e^* \succeq 0$, every eigenvalue and hence $s \mapsto \operatorname{Tr} F(L_{J(e,s),L})$ is nondecreasing. Together with boundedness of the density this implies

$$\int \rho(s) \partial_s \operatorname{Tr} F(L_{J(e,s),L}) ds \leq \rho_\infty [\operatorname{Tr} F(L_{j_+}) - \operatorname{Tr} F(L_{j_-})].$$

The endpoint matrices differ by the positive rank-one operator $(j_+ - j_-) b_e b_e^*$. Rank-one interlacing and the fact that the range of F has length ℓ give

$$0 \leq \operatorname{Tr} F(L_{j_+}) - \operatorname{Tr} F(L_{j_-}) \leq \ell.$$

Summing over the dL^d positive-coordinate bonds proves (43). Finally, $L_{J,L} \succeq j_- L_0$ and the lattice Weyl count give (44). $\square$

The count theorem is the finite discrete analogue of divergence-form Wegner estimates, for which eigenvalue lifting or gradient unique continuation is needed in the continuum [7, 8]. Here the exact bond covering $L_0 = \sum_e b_e b_e^*$ and rank-one interlacing make the proof self-contained.

It is nevertheless not a drift-weighted Wegner theorem. The modal estimate is only

$$\frac{1}{L^d} \frac{|\langle g_L, \psi_k \rangle|^2}{\lambda_k} \leq j_+ |A|^2, \quad (45)$$

not $O(L^{-d})$. A coherent soft mode can therefore carry an order-one fraction of the whole H^{-1} source mass.

Proposition 4.2 (Normalized eigenvalue count does not control drift weight). *On the N -cycle, let*

$$j_{e,N} = 1 + \varepsilon \cos(2\pi e/N), \quad 0 < \varepsilon < 1, \quad (46)$$

and take $a_e = 1$. There are constants $c_\varepsilon, C_\varepsilon > 0$ such that, for all sufficiently large N ,

$$\boxed{\mu_N((0, C_\varepsilon N^{-2}]) \geq c_\varepsilon, \quad \frac{1}{N} \mathcal{N}_N((0, C_\varepsilon N^{-2}]) \rightarrow 0.} \quad (47)$$

In particular, no realization-wise inequality $\mu_{L,J}(I) \leq CL^{-d} \mathcal{N}_L(I)$ can hold uniformly over bounded elliptic conductances.

Proof. The total mass is the arithmetic-minus-harmonic conductance,

$$\mu_N((0, \infty)) = 1 - \left(\frac{1}{N} \sum_e j_{e,N}^{-1} \right)^{-1} \rightarrow 1 - \sqrt{1 - \varepsilon^2} > 0.$$

The first moment is explicit:

$$\int \lambda d\mu_N(\lambda) = \frac{1}{N} \|g_N\|^2 = 2\varepsilon^2 \sin^2(\pi/N) \leq 2\pi^2 \varepsilon^2 N^{-2}.$$

Markov's inequality for the measure μ_N therefore places a fixed positive fraction of its mass below $C_\varepsilon N^{-2}$ once C_ε is large enough. On the other hand, $L_{J_N} \succeq (1 - \varepsilon)L_0$, and the clean cycle has only $O_{\varepsilon,C}(1)$ eigenvalues below CN^{-2} . Dividing by N proves (47). $\square$

This proposition prevents a shortcut through the ordinary IDS. Recent uniform concentration results for normalized eigenvalue counting functions remain count-level statements [9]; the needed theorem must control current overlap with the low-energy subspace.

Route II: randomized cutoffs. Let Θ_δ be independent of the conductances and exponentially distributed with mean δ .

Theorem 4.3 (Abel tail as an exactly randomized sharp cutoff). *For every realization,*

$$\boxed{\mathbb{E}_\Theta T_L(\Theta_\delta) = \mathfrak{A}_L(\delta)}, \quad (48)$$

$$\boxed{\mathbb{E}_\Theta T_L(\Theta_\delta)^2 = \iint e^{-\max\{\lambda, \mu\}/\delta} d\mu_{L,J}(\lambda) d\mu_{L,J}(\mu) \leq \mathfrak{A}_L(2\delta)^2}. \quad (49)$$

Consequently, for $0 < \delta \leq j_+/2$,

$$\mathbb{E}_{J,\Theta} T_L(\Theta_\delta)^2 \leq C j_+^2 |A|^4 \left[\left(\frac{\delta}{j_+} \right)^d + L^{-d} \left(\frac{\delta}{j_+} \right)^{d/2} \right]. \quad (50)$$

The disorder variance of the conditional mean is exactly the Abel variance:

$$\text{Var}_J(\mathbb{E}_\Theta[T_L(\Theta_\delta) \mid J]) = \text{Var}_J \mathfrak{A}_L(\delta), \quad (51)$$

and hence obeys (23).

Proof. The survival probability is $\mathbb{P}(\Theta_\delta \geq \lambda) = e^{-\lambda/\delta}$, so Tonelli gives (48). Applying the same identity twice gives the double integral in (49). Since $\max\{\lambda, \mu\} \geq (\lambda + \mu)/2$, its integrand is bounded by $e^{-\lambda/(2\delta)} e^{-\mu/(2\delta)}$. The final claims follow from (13) and (23) at scale 2δ . $\square$

Thus the Abel theorem is already an optimal centered disorder theorem for a precisely defined randomized-frequency protocol. It does not, however, recover a deterministic threshold. The total-variance identity

$$\text{Var}_{J,\Theta} T_L(\Theta_\delta) = \text{Var}_J \mathfrak{A}_L(\delta) + \mathbb{E}_J \text{Var}_\Theta(T_L(\Theta_\delta) \mid J) \quad (52)$$

contains threshold noise. For the one-atom measure $\mu = m\delta_{\{\delta\}}$, that term is $m^2 e^{-1}(1 - e^{-1})$; if $m \asymp \delta^{d/2}$, it is of order δ^d , not $L^{-d}\delta^{d/2}$. Broad randomization therefore cannot by itself prove the fixed-cutoff CLT scale. In fact this obstruction applies to every positive heat-kernel mixture.

Proposition 4.4 (No narrow cutoff from a positive heat mixture). *Let a randomized-threshold survival function have the completely monotone form*

$$s(\lambda) = \int_0^\infty e^{-t\lambda} \pi(dt), \quad \pi((0, \infty)) = 1. \quad (53)$$

If the associated threshold Θ has finite second moment, then

$$\boxed{\text{Var } \Theta \geq (\mathbb{E}\Theta)^2}. \quad (54)$$

Thus no family of positive heat mixtures can concentrate its threshold in a relative window $o(1)$ around its mean.

Proof. Conditionally on t , let Θ be exponential with rate t . Then (53) is its survival function and

$$\mathbb{E}\Theta = \int t^{-1} \pi(dt), \quad \mathbb{E}\Theta^2 = 2 \int t^{-2} \pi(dt).$$

Cauchy–Schwarz gives $\int t^{-2} d\pi \geq (\int t^{-1} d\pi)^2$, proving (54). $\square$

The proposition explains why positivity and narrowness conflict in a pure heat-mixture strategy. Erlang or gamma thresholds with large shape do concentrate, but their survival functions contain polynomial times exponential multipliers rather than positive mixtures of exponentials. Exploiting them would require uniform high-order semigroup-derivative estimates.

For a narrow randomization, let $U_{\delta,\eta}$ be uniform on $[\delta, (1+\eta)\delta]$ and put

$$\bar{T}_{L,\eta}(\delta) = \mathbb{E}_U[T_L(U_{\delta,\eta}) \mid J]. \quad (55)$$

Its spectral multiplier equals one below δ , decreases linearly across the shell, and vanishes above $(1+\eta)\delta$. Hence, samplewise,

$$\boxed{0 \leq \bar{T}_{L,\eta}(\delta) - T_L(\delta) \leq S_L(\delta, \eta)}, \quad (56)$$

$$\boxed{\text{Var}_U(T_L(U_{\delta,\eta}) \mid J) \leq \frac{1}{4} S_L(\delta, \eta)^2}. \quad (57)$$

Narrow randomization gives the desired smooth/sharp sandwich exactly, but it also shows that the shell modulus cannot be avoided.

Route III: the integrated spectral variable. Define

$$K_L(r) = \int_0^r T_L(s) ds = \int (r - \lambda)_+ d\mu_{L,J}(\lambda). \quad (58)$$

The multiplier is continuous at $\lambda = r$, and

$$\boxed{\frac{K_L(r+h) - K_L(r)}{h} = \frac{1}{h} \int_r^{r+h} T_L(s) ds.} \quad (59)$$

For $h = \eta r$, the right-hand side is $\bar{T}_{L,\eta}(r)$, so

$$0 \leq \frac{K_L(r+h) - K_L(r)}{h} - T_L(r) \leq S_L(r, h/r). \quad (60)$$

Thus the integrated-variable and randomized-cutoff routes are the same regularization in two languages. Although pointwise shell control remains open, its logarithmic-energy average closes exactly.

Theorem 4.5 (Exact logarithmic shell identity and averaged shell bound). *For every finite positive measure μ on $(0, \infty)$, set $S_\mu(r, \eta) = \mu((r, (1 + \eta)r])$. Then*

$$\int_0^\infty S_\mu(r, \eta)^2 \frac{dr}{r} = \iint [\log(1 + \eta) - |\log(\lambda/\lambda')|]_+ \mu(d\lambda) \mu(d\lambda'). \quad (61)$$

Consequently, for the drift measure (41), $0 < \eta \leq 1$, and $0 < R \leq j_+/(1 + \eta)$,

$$\int_0^R \mathbb{E} S_L(r, \eta)^2 \frac{dr}{r} \leq C \eta j_+^2 |A|^4 \left[\left(\frac{R}{j_+} \right)^d + L^{-d} \left(\frac{R}{j_+} \right)^{d/2} \right]. \quad (62)$$

The same estimate holds for the integrated difference quotient:

$$\int_0^R \mathbb{E} \left| \frac{K_L(r + \eta r) - K_L(r)}{\eta r} - T_L(r) \right|^2 \frac{dr}{r} \leq \text{right-hand side of (62)}. \quad (63)$$

Proof. For fixed λ , the condition $r < \lambda \leq (1 + \eta)r$ means that $\log r$ lies in an interval of length $\log(1 + \eta)$. The intersection of the two intervals associated with λ and λ' has length $[\log(1 + \eta) - |\log(\lambda/\lambda')|]_+$. Tonelli proves (61). Restricting $r \leq R$ and discarding the intersection geometry gives the deterministic bound

$$\int_0^R S_L(r, \eta)^2 \frac{dr}{r} \leq \log(1 + \eta) T_L((1 + \eta)R)^2. \quad (64)$$

Take expectations, use (29), and absorb $1 + \eta \leq 2$ and $\log(1 + \eta) \leq \eta$. Finally, (60) proves (63). $\square$

Lemma 4.6 (Pointwise maximalization at the unavoidable floor). *For every finite positive measure μ , every $r > 0$, and every $\eta > 0$, put $h = \log(1 + \eta)$ and $\eta^{(2)} = (1 + \eta)^2 - 1$. Then*

$$S_\mu(r, \eta)^2 \leq \frac{1}{h} \int_{r/(1+\eta)}^r S_\mu(s, \eta^{(2)})^2 \frac{ds}{s}. \quad (65)$$

Hence, for $0 < \eta \leq \sqrt{2} - 1$ and $r \leq j_+/(1 + \eta)^2$,

$$\mathbb{E} S_L(r, \eta)^2 \leq C j_+^2 |A|^4 \left[(r/j_+)^d + L^{-d} (r/j_+)^{d/2} \right]. \quad (66)$$

There is no exceptional cutoff in (66), but the bound has no vanishing power of η .

Proof. Push μ forward by $\lambda \mapsto x = \log \lambda$. If $y \in [x - h, x]$, then the interval $(x, x + h]$ is contained in $(y, y + 2h]$. Thus its mass squared is bounded by the average of the wider-shell mass squared over $y \in [x - h, x]$, which is (65). Take expectations and apply (62) with relative width $\eta^{(2)} \leq 1$ and upper endpoint r . Since $\eta^{(2)}/\log(1 + \eta)$ is uniformly bounded on $(0, \sqrt{2} - 1]$, this gives (66). $\square$

The lemma is a direct conclusion obtainable from positivity and the current second-moment input. In the bounded-mode window its right-hand side is the target variance scale by theorem 2.7; in the mesoscopic window the first term is larger by m_{eff} . The acoustic crossing model below shows why a volume-uniform improvement by a factor η^α cannot be expected at arbitrarily thin prescribed shells.

In particular, if R_* is independent and log-uniform on $[R/2, R]$, then

$$\mathbb{E}_{J, R_*} S_L(R_*, \eta)^2 \leq C \eta j_+^2 |A|^4 \left[(R/j_+)^d + L^{-d} (R/j_+)^{d/2} \right]. \quad (67)$$

Thus the desired shell modulus is already true for a typical cutoff in every dyadic energy window. What is missing is a bound uniform at a prescribed cutoff.

Corollary 4.7 (Exceptional set of spectral cutoffs). *Fix $0 < \beta < 1$, $0 < \eta \leq 1$, and put*

$$Q_L(R) = j_+^2 |A|^4 \left[(R/j_+)^d + L^{-d} (R/j_+)^{d/2} \right]. \quad (68)$$

There is a constant C such that the bad set

$$\mathcal{B}_{L,R}(\eta, \beta) = \left\{ r \in [R/2, R] : \mathbb{E} S_L(r, \eta)^2 > C \eta^\beta Q_L(R) \right\} \quad (69)$$

obeys

$$\boxed{\int_{\mathcal{B}_{L,R}(\eta, \beta)} \frac{dr}{r} \leq C \eta^{1-\beta}.} \quad (70)$$

Consequently, let (L_k) be any deterministic volume sequence and let $\eta_k \downarrow 0$ satisfy $\sum_k \eta_k^{1-\beta} < \infty$. Then for logarithmically almost every $r \in [R/2, R]$,

$$\mathbb{E} S_{L_k}(r, \eta_k)^2 \leq C \eta_k^\beta Q_{L_k}(R) \quad (71)$$

for all sufficiently large k . The same conclusion holds for the L^2 error between $T_{L_k}(r)$ and the collapsing integrated difference quotient with $h_k = \eta_k r$.

Proof. Restrict (62) to $[R/2, R]$ and apply Markov's inequality with respect to logarithmic measure. This gives (70). The second statement follows from the Borel–Cantelli lemma on the finite measure space $([R/2, R], dr/r)$; no independence in k is needed. The final claim uses (60). $\square$

Thus the pointwise shell problem is already solved outside a quantitatively small deterministic set of cutoff energies, and along arbitrary volume sequences it is solved for almost every energy. The remaining theorem asks for a bound at a prescribed cutoff with no exceptional set.

Order the dL^d conductances, let $(\mathcal{F}_m)$ be their reveal filtration, and put

$$Y_{r,h} = \frac{K_L(r+h) - K_L(r)}{h}, \quad D_m(r, h) = \mathbb{E}[Y_{r,h} \mid \mathcal{F}_m] - \mathbb{E}[Y_{r,h} \mid \mathcal{F}_{m-1}]. \quad (72)$$

The exact Doob identity is

$$\text{Var } Y_{r,h} = \sum_{m=1}^{dL^d} \mathbb{E} |D_m(r, h)|^2. \quad (73)$$

5 Pointwise rank-one geometry, participation coarea, and the correct martingale target

The logarithmic shell theorem leaves open whether its exceptional set can be removed by a pointwise maximal estimate. We now resolve the finite-dimensional geometry of that question. The first conclusion is favorable: varying one bond cannot produce multiple crossings of a fixed cutoff. The second is a warning: the resulting one-bond Efron–Stein surface term is generally stronger than the centered theorem and can lose a square-root volume factor at an individual acoustic eigenvalue. This identifies direct Doob smoothing in *disorder space*, rather than a volume-uniform spectral shell modulus, as the correct remaining route.

Write $N = L^d$, $b_e = B\mathbf{e}_e$, and, after deleting the e th bond,

$$L_e^{(0)} = \sum_{f \neq e} j_f b_f b_f^*, \quad g_e^{(0)} = \sum_{f \neq e} j_f a_f b_f. \quad (74)$$

For $t \in [j_-, j_+]$ put

$$L_e(t) = L_e^{(0)} + tb_e b_e^*, \quad g_e(t) = g_e^{(0)} + ta_e b_e. \quad (75)$$

All operators below are restricted to $\mathbf{1}^\perp$.

Lemma 5.1 (Single-crossing lemma). *For $t_1 < t_2$ and every $r > 0$,*

$$0 \leq \text{tr } \mathbf{1}_{(0,r]}(L_e(t_1)) - \text{tr } \mathbf{1}_{(0,r]}(L_e(t_2)) \leq 1. \quad (76)$$

Thus a complete one-bond path crosses a prescribed positive energy at most once, counting multiplicity.

Proof. The perturbation $L_e(t_2) - L_e(t_1) = (t_2 - t_1)b_e b_e^*$ is positive and rank one. Monotonicity and rank-one Cauchy interlacing imply that the counting function below r can only decrease and can decrease by at most one. $\square$

The unique crossing has explicit leave-one-edge coordinates. For $r \notin \sigma(L_e^{(0)}|_{\mathbf{1}^\perp})$, define

$$m_e(r) = \langle b_e, (L_e^{(0)} - r)^{-1} b_e \rangle, \quad \dot{m}_e(r) = \langle b_e, (L_e^{(0)} - r)^{-2} b_e \rangle, \quad (77)$$

$$h_e(r) = \langle b_e, (L_e^{(0)} - r)^{-1} g_e^{(0)} \rangle. \quad (78)$$

Proposition 5.2 (Krein coordinates for the unique crossing). *A crossing occurs in the ellipticity interval precisely when*

$$t_e(r) = -\frac{1}{m_e(r)} \in (j_-, j_+). \quad (79)$$

Because the admissible conductance is positive, such a crossing necessarily has $m_e(r) < 0$; equivalently, r lies above at least one eigenvalue of the deleted-bond operator on $\mathbf{1}^\perp$. At that crossing, a normalized eigenvector, the eigenvalue speed, and the jump of the sharp tail are

$$\psi_e(r) = \frac{(L_e^{(0)} - r)^{-1} b_e}{\sqrt{\dot{m}_e(r)}}, \quad (80)$$

$$\partial_t \lambda_e(t)|_{t=t_e(r)} = \frac{m_e(r)^2}{\dot{m}_e(r)}, \quad (81)$$

$$q_e(r) = \frac{1}{Nr} |\langle g_e(t_e(r)), \psi_e(r) \rangle|^2 = \frac{|h_e(r) - a_e|^2}{Nr \dot{m}_e(r)}. \quad (82)$$

As t increases through $t_e(r)$, the function $T_L(r; J_{\setminus e}, t)$ has a single downward jump of size $q_e(r)$. Moreover, for every $z \notin \sigma(L_e(t))$,

$$\begin{aligned} & \langle g_e(t), (L_e(t) - z)^{-1} g_e(t) \rangle \\ &= \langle g_e^{(0)}, (L_e^{(0)} - z)^{-1} g_e^{(0)} \rangle + \frac{t(a_e^2 m_e(z)t + 2a_e h_e(z) - h_e(z)^2)}{1 + t m_e(z)}. \end{aligned} \quad (83)$$

Proof. The matrix determinant lemma gives $\det(L_e(t) - r) = \det(L_e^{(0)} - r)(1 + t m_e(r))$, which proves (79). Applying $L_e(t) - r$ to $(L_e^{(0)} - r)^{-1} b_e$ gives $(1 + t m_e(r))b_e$, and normalization yields (80). Hellmann–Feynman gives $\partial_t \lambda = |\langle b_e, \psi_e \rangle|^2 = m_e^2 / \dot{m}_e$. At $t = -1/m_e$,

$$\langle g_e(t), \psi_e \rangle = \frac{h_e + ta_e m_e}{\sqrt{\dot{m}_e}} = \frac{h_e - a_e}{\sqrt{\dot{m}_e}},$$

which proves (82). Finally, the Sherman–Morrison formula, together with the affine dependence of $g_e(t)$, gives (83) by direct expansion. $\square$

Let (λ_k, ψ_k) denote the positive eigenpairs of $L_{J,L}$ and set

$$q_k(J) = \frac{1}{N\lambda_k} |\langle g_L, \psi_k \rangle|^2, \quad \Pi_L(I) = \mathbb{E} \sum_k q_k(J)^2 \mathbf{1}_I(\lambda_k). \quad (84)$$

The measure Π_L is the diagonal, or inverse-participation, part of the second moment of the drift spectral measure. It is exactly the coefficient of the linear small-shell term; off-diagonal pairs contribute only at higher order when the disorder-averaged spectral law is regular.

Theorem 5.3 (Participation coarea formula). *Assume that the edge law has density ρ on $[j_-, j_+]$ and that the positive spectrum is simple almost surely. Then Π_L is absolutely continuous on $(0, \infty)$. For almost every $r > 0$, its density $\pi_L(r) = d\Pi_L/dr$ satisfies*

$$r\pi_L(r) = \sum_e \mathbb{E}_{\setminus e} \left[t_e(r) \rho(t_e(r)) q_e(r)^2 \mathbf{1}_{\{t_e(r) \in (j_-, j_+)\}} \right]. \quad (85)$$

Equivalently, the explicit leave-one-edge Green-function target is

$$r\pi_L(r) = \frac{1}{N^2 r^2} \sum_e \mathbb{E}_{\setminus e} \left[t_e(r) \rho(t_e(r)) \frac{|h_e(r) - a_e|^4}{\tilde{m}_e(r)^2} \mathbf{1}_{\{t_e(r) \in (j_-, j_+)\}} \right]. \quad (86)$$

In particular,

$$\mathbb{E} \sum_{r < \lambda_k \leq (1+\eta)r} q_k^2 = \int_r^{(1+\eta)r} \pi_L(s) ds. \quad (87)$$

Proof. Every positive eigenvalue is homogeneous of degree one in the conductances, so Euler's identity and Hellmann–Feynman give

$$\sum_e \frac{j_e \partial_{j_e} \lambda_k}{\lambda_k} = 1. \quad (88)$$

For a compactly supported test function f ,

$$\int f(r) \Pi_L(dr) = \sum_e \mathbb{E} \sum_k \frac{j_e \partial_{j_e} \lambda_k}{\lambda_k} q_k^2 f(\lambda_k).$$

Condition on all bonds except $j_e = t$. The eigenvalue branches are monotone, and theorem 5.1 says that at a fixed r there is at most one crossing over the full t interval. Changing variables from t to the crossing eigenvalue cancels $\partial_t \lambda_k$ and gives

$$\int f(r) \frac{1}{r} \mathbb{E}_{\setminus e} \left[t_e(r) \rho(t_e(r)) q_e(r)^2 \mathbf{1}_{\{t_e(r) \in (j_-, j_+)\}} \right] dr.$$

Summing over e proves (85); (86) follows from (82). $\square$

The coarea formula also gives the exact one-bond surface term in the product variance inequality. Let $F_\beta(t) = \beta([j_-, t])$, and, conditionally on $J_{\setminus e}$, put

$$H_e(t) = T_L(r; J_{\setminus e}, t), \quad \tilde{H}_e(t) = H_e(t) + q_e(r) \mathbf{1}_{\{t \geq t_e(r)\}}, \quad (89)$$

with $q_e = 0$ when there is no crossing. The function $\tilde{H}_e$ is continuous and piecewise analytic.

Theorem 5.4 (Exact Efron–Stein surface decomposition). *Suppose the one-edge law satisfies the derivative Poincaré inequality (22). Define*

$$\mathcal{R}_L(r) = \sum_e \mathbb{E}_{\setminus e} \int \left| \partial_t \tilde{H}_e(t) \right|^2 \beta(dt), \quad (90)$$

$$\mathcal{J}_L(r) = \sum_e \mathbb{E}_{\setminus e} \left[q_e(r)^2 F_\beta(t_e(r))(1 - F_\beta(t_e(r))) \mathbf{1}_{\{t_e(r) \in (j_-, j_+)\}} \right]. \quad (91)$$

Then

$$\boxed{\text{Var } T_L(r) \leq 2C_\beta \mathcal{R}_L(r) + 2\mathcal{J}_L(r).} \quad (92)$$

If

$$\kappa_\beta = \text{ess sup}_{j_- < t < j_+} \frac{F_\beta(t)(1 - F_\beta(t))}{t\rho(t)} < \infty, \quad (93)$$

then

$$\boxed{\mathcal{J}_L(r) \leq \kappa_\beta r \pi_L(r).} \quad (94)$$

For the uniform law on $[j_-, j_+]$, $\kappa_\beta \leq (j_+ - j_-)/(4j_-)$.

Proof. Tensorization gives $\text{Var } T_L(r) \leq \sum_e \mathbb{E}_{\setminus e} \text{Var}_\beta H_e$. Since $H_e = \tilde{H}_e - q_e \mathbf{1}_{\{t \geq t_e\}}$,

$$\text{Var}_\beta H_e \leq 2 \text{Var}_\beta \tilde{H}_e + 2q_e^2 F_\beta(t_e)(1 - F_\beta(t_e)).$$

Apply the one-dimensional Poincaré inequality to $\tilde{H}_e$ and sum. Equation (94) follows term by term from (85) and (93). $\square$

This theorem removes the previously stated “multiple crossing” obstruction: there is only one crossing channel. It also shows that the naive pointwise target

$$r\pi_L(r) \stackrel{?}{\lesssim} j_+^2 |A|^4 N^{-1} (r/j_+)^{d/2} \quad (95)$$

is an Efron–Stein sufficient condition, not a necessary consequence of the centered variance theorem. In fact, at a cutoff passing through an individual acoustic eigenvalue, (95) is not the scale predicted by eigenvalue central-limit fluctuations.

Proposition 5.5 (Acoustic-CLT obstruction to a uniform maximal shell modulus). *Let $N \rightarrow \infty$, let $r_N > 0$, and consider a single positive modal contribution*

$$\lambda_N = r_N \exp(Z_N/\sqrt{N}), \quad q_N = Q_N/N. \quad (96)$$

Assume $(Z_N, Q_N) \Rightarrow (Z, Q)$, the family Q_N^2 is uniformly integrable, and for some $a > 0$,

$$\mathbb{E} \left[Q^2 \mathbf{1}_{\{0 < Z \leq a\}} \right] > 0. \quad (97)$$

Put $\eta_N = \exp(a/\sqrt{N}) - 1$. Then the one-mode shell obeys

$$N^2 \mathbb{E} \left[q_N^2 \mathbf{1}_{\{r_N < \lambda_N \leq (1+\eta_N)r_N\}} \right] \longrightarrow \mathbb{E} \left[Q^2 \mathbf{1}_{\{0 < Z \leq a\}} \right] > 0. \quad (98)$$

Consequently, for every $\alpha > 0$, no volume-uniform estimate of the form

$$\mathbb{E} S_N(r_N, \eta)^2 \leq C \eta^\alpha N^{-2} \quad (99)$$

can hold at all η , even though the centered one-mode tail has variance of order N^{-2} . In the explicit Gaussian case $Z \sim \mathcal{N}(0, 1)$ and $Q \equiv q_*$,

$$r_N \pi_N(r_N) = \frac{q_*^2}{\sqrt{2\pi}} N^{-3/2}, \quad \text{Var} \left[\frac{q_*}{N} \mathbf{1}_{\{Z \leq 0\}} \right] = \frac{q_*^2}{4} N^{-2}. \quad (100)$$

Thus the pointwise one-bond surface density can be larger than the centered variance scale by $\sqrt{N}$.

Proof. The shell event is exactly $\{0 < Z_N \leq a\}$. Uniform integrability and weak convergence give (98). Since $\eta_N^\alpha \sim a^\alpha N^{-\alpha/2}$, (99) would force the positive limit in (98) to vanish. The Gaussian formulas follow from the density of $Z/\sqrt{N}$ at the origin and from the variance of a fair Bernoulli variable. $\square$

The proposition is an exact surrogate statement, not an assertion that a specific torus eigenpair already satisfies the required joint limit. It is, however, the scaling expected from quantitative eigenvalue fluctuation theory for isolated homogenized elliptic modes [10]; large-scale acoustic dispersive estimates give complementary context [11]. The finite-torus audit in the supplementary package finds $\text{sd}(\log \lambda) \simeq N^{-1/2}$, $r\pi_L(r) \simeq N^{-3/2}$, and $\text{Var } T_L(r) \simeq N^{-2}$ across the first acoustic multiplet in two dimensions. These diagnostics do not enter the proof.

The correct closure statement must therefore allow a volume-dependent spectral smoothing scale, or avoid spectral smoothing altogether.

Theorem 5.6 (Diagonal integrated-martingale criterion). *Let r_L lie in the diffusive window and define*

$$V_L(r_L) = j_+^2 |A|^4 N^{-1} (r_L/j_+)^{d/2}. \quad (101)$$

Suppose there is a sequence $h_L \downarrow 0$ such that

$$\mathbb{E} S_L(r_L, h_L/r_L)^2 = o(V_L(r_L)), \quad (102)$$

$$\sum_m \mathbb{E} |D_m(r_L, h_L)|^2 \leq C V_L(r_L). \quad (103)$$

Then

$$\text{Var } T_L(r_L) \leq (C + o(1)) V_L(r_L). \quad (104)$$

No volume-uniform limit $h \downarrow 0$ is required.

Proof. By (60), $\|Y_{r_L, h_L} - T_L(r_L)\|_2 = o(V_L^{1/2})$. The exact Doob identity (73) and the triangle inequality for centered L^2 norms then give (104). $\square$

There is an even cleaner fixed-cutoff formulation. Define the sharp Doob increments

$$D_m^\sharp(r) = \mathbb{E}[T_L(r) \mid \mathcal{F}_m] - \mathbb{E}[T_L(r) \mid \mathcal{F}_{m-1}]. \quad (105)$$

Then

$$\boxed{\text{Var } T_L(r) = \sum_m \mathbb{E} |D_m^\sharp(r)|^2.} \quad (106)$$

The full centered theorem is therefore equivalent to a direct pointwise bound on this quadratic variation. Unlike Efron–Stein, the Doob increments average each crossing over all unrevealed conductances and can capture the acoustic central-limit smoothing responsible for the missing $N^{-1/2}$ factor.

A useful exact representation of that smoothing is provided by the reversible noise semigroup on disorder space. Let

$$\mathcal{G} = \sum_e (\mathbb{E}_e - \mathbb{I}), \quad P_t = e^{t\mathcal{G}}, \quad (107)$$

where $\mathbb{E}_e$ averages only the e th conductance. The product law is reversible and

$$\boxed{\text{Var } F = 2 \int_0^\infty \sum_e \mathbb{E} \text{Var}_e(P_t F) dt.} \quad (108)$$

Thus it is sufficient to prove, at the fixed physical cutoff,

$$\sum_e \mathbb{E} \text{Var}_e(P_t T_L(r)) \leq C V_L(r) \vartheta(t), \quad \int_0^\infty \vartheta(t) dt < \infty. \quad (109)$$

This route never differentiates the spectral projector in r and has no exceptional cutoff set.

The expected short-time singularity in (109) is integrable and is already visible in the exact Gaussian half-space model. If $X \in \mathbb{R}^M$ is standard Gaussian, $Z = M^{-1/2} \sum_i X_i$, and $F = q \mathbf{1}_{\{Z \leq 0\}}$, then the Ornstein–Uhlenbeck semigroup satisfies

$$P_t F = q \Phi \left(-\frac{e^{-t} Z}{\sqrt{1 - e^{-2t}}} \right). \quad (110)$$

Writing $a_t = e^{-t}/\sqrt{1 - e^{-2t}}$ gives

$$\mathbb{E} |\nabla P_t F|^2 = \frac{q^2 a_t^2}{2\pi \sqrt{1 + 2a_t^2}} \sim \frac{q^2}{4\pi \sqrt{t}} \quad (t \downarrow 0), \quad (111)$$

while it decays exponentially for large t , and exactly

$$2 \int_0^\infty \mathbb{E} |\nabla P_t F|^2 dt = \frac{q^2}{4} = \text{Var } F. \quad (112)$$

The $t^{-1/2}$ boundary layer is the disorder-space replacement for the false volume-uniform shell modulus. The new sharp target is to prove the analogue of (111) for the low-energy current-weighted random-conductance functional, using the coarea coordinates (86) together with conditional homogenization of the unrevealed environment.

Problem 5.7 (Pointwise disorder-noise energy). Prove the fixed-cutoff estimate

$$\int_0^\infty \sum_e \mathbb{E} \text{Var}_e(P_t T_L(r)) dt \leq C j_+^2 |A|^4 N^{-1} (r/j_+)^{d/2} \quad (113)$$

uniformly for $cL^{-2} \leq r/j_+ \leq \delta_0$. A stronger, scale-resolved version would establish (109) with $\vartheta(t) \lesssim t^{-1/2}$ for $t \leq 1$ and exponential decay for $t \geq 1$. The coarea theorem identifies the exact crossing surface, while the acoustic obstruction shows why the unrevealed-environment smoothing in P_t or in the direct Doob martingale is essential. Proving (113) closes the centered nonlinear theorem without any exceptional set of physical cutoff energies.

6 Numerical audit

The Paper II archive retains seven independent programs. They verify the Abel heat and parabolic-energy identities, the exact Duhamel bond derivative, the Born Fourier formulas and Hanson–Wright normalization, the relative Wegner and randomized-cutoff identities, the logarithmic shell formula, the single-crossing/Krein/coarea relations, and the Gaussian and Bernoulli disorder-noise boundary layers. The exact Abel Duhamel/spectral derivative discrepancy is 2.17×10^{-19} , the finite-difference sensitivity error is 1.05×10^{-12} , the one-crossing eigenvalue error is 6.66×10^{-16} , and the exact Gaussian noise-energy integral closes to 7.92×10^{-14} . Monte Carlo coarea and low-mode scaling tables are explicitly labeled diagnostics rather than proof inputs.

7 Theorem boundary

The positive Abel envelope has the predicted effective-mode variance under the stated one-bond Poincaré and period-uniform GNO hypotheses. The physical sharp tail inherits a one-sided high-probability bound. At bounded effective-mode count, the target variance scale follows from the uncentered second moment; genuine centering remains necessary in the mesoscopic window. The Born sharp observable has a complete centered concentration theorem.

The three-route audit closes several tempting shortcuts. Counts alone do not control drift weight; positive heat mixtures cannot create a narrow threshold; logarithmic shell averaging does not imply a vanishing pointwise shell modulus; and a single bond creates at most one cutoff crossing. The rank-one and coarea formulas isolate the exact surface term that direct Efron–Stein overcounts near an acoustic mode.

Problem 7.1 (Fixed-cutoff disorder-noise theorem). Prove, uniformly for $cL^{-2} \leq r/j_+ \leq \delta_0$ in the mesoscopic regime $m_{\text{eff}}(L, r) \rightarrow \infty$,

$$2 \int_0^\infty \sum_e \mathbb{E} \text{Var}_e(P_t T_L(r)) dt \leq C j_+^2 |A|^4 L^{-d} (r/j_+)^{d/2}.$$

A scale-resolved estimate with an integrable $t^{-1/2}$ short-time profile would close the centered nonlinear theorem at every prescribed cutoff without an exceptional energy set and without a false uniform maximal-shell hypothesis.

8 Conclusion

The smooth and weak-disorder sectors now have complete fluctuation theorems at the effective diffusive-mode scale. For the nonlinear sharp cutoff, the obstruction is not an unidentified eigenvalue crossing: rank-one geometry makes the crossing unique and explicit. Nor is it a small exceptional set of physical frequencies: pointwise maximalization reaches the unavoidable uncentered floor at every cutoff. The missing mechanism is averaging in disorder space. The exact product-noise representation identifies the final theorem as an integrable fixed-cutoff quadratic-variation estimate, with the acoustic half-space model predicting the correct $t^{-1/2}$ boundary layer.

A Periodic gradient propagator: definitions, imported estimates, and proof

Normalize $j_+ = 1$ and put $\lambda_* = j_-/j_+$. For a fixed periodic environment J , define the torus heat kernel by

$$G_L(t, J, x, y) = (e^{-tL_{J,L}}\delta_y)(x), \quad x, y \in \Lambda_L. \quad (114)$$

It solves $\partial_t G_L + L_{J,L} G_L = 0$ with initial datum δ_y and conserves total mass. Gradients annihilate its constant long-time component, so no mean-zero subtraction is required in the estimates below. Let

$$\omega_{L,t}(y) = \left(1 + \frac{\text{dist}(y, L\mathbb{Z}^d)^2}{1+t}\right)^{1/2}, \quad (115)$$

where $\text{dist}(y, L\mathbb{Z}^d)$ is the Euclidean distance of a representative of y to the nearest period point.

We use two inputs from the long version of Gloria–Neukamm–Otto [3]; the corresponding quantitative homogenization framework is published in [4]. The period-uniform statements below are taken from the numbered results of the long version.

For completeness, we spell out the periodic vertical hypothesis. Let $Q_L = \{0, \dots, L-1\}^d$, regard the d outgoing coefficients at a site as one coefficient block, and set

$$\partial_{L,y} F = F - \mathbb{E}[F \mid \{j(x) : x \in Q_L, x \neq y\}]. \quad (116)$$

The condition $\text{SG}_L(\rho)$ is

$$\text{Var } F \leq \rho^{-1} \sum_{y \in Q_L} \mathbb{E} |\partial_{L,y} F|^2 \quad (117)$$

for every square-integrable mean-zero F . The product law of independent site blocks satisfies this inequality with a constant independent of L by tensorization. For a vector-valued random field ξ , write

$$\|\xi\|_{\ell_y^1 L^{2p}} := \sum_{y \in Q_L} \|\partial_{L,y} \xi\|_{L^{2p}(\Omega_L)}. \quad (118)$$

Theorem A.1 (Imported divergence-semigroup estimate). *Assume the stationary L -periodic ensemble satisfies $\text{SG}_L(\rho)$ with ρ independent of L . Theorem 2 under alternative (69b) supplies $p_0 = p_0(d, \lambda_*) < \infty$ such that, for $p \geq p_0$ and $u(t) = e^{-t\mathcal{L}_L} D^* \xi$,*

$$\|u(t)\|_{L^{2p}(\Omega_L)} \leq C_p (1+t)^{-d/4-1/2} \|\xi\|_{\ell_y^1 L^{2p}}, \quad (119)$$

where $C_p = C(p, d, \lambda_*, \rho)$ has no dependence on L . Lower finite moments follow by Jensen. For the local drift $d_A = D^*(j(0)A)$, Remark 11 and equation (70) give

$$\left\| e^{-t\mathcal{L}_L} d_A \right\|_{L^p(\Omega_L)} \leq C_p |A| (1+t)^{-d/4-1/2}. \quad (120)$$

Theorem A.2 (Imported periodic Green-gradient estimate). *Theorem 3(b), with the weight in its equation (74), supplies an exponent $q > 1$ depending only on d, λ_* such that for every $\alpha < \infty$,*

$$\left(\sum_{y \in \Lambda_L} [\omega_{L,t}(y)^\alpha |\nabla_y G_L(t, J, y, 0)|]^{2q} \right)^{1/(2q)} \leq C_{\alpha,q} (1+t)^{-d/2-1/2+d/(4q)} e^{-ct/L^2}. \quad (121)$$

The estimate is deterministic in J , uniform over all L -periodic coefficients in $[\lambda_*, 1]$, and uniform in L .

The product torus law satisfies $\text{SG}_L(\rho)$ by tensorization of its one-block spectral gap. We now derive the lemma used in the main text. Let $F_y(J) = F(\tau_y J)$ and set $r = 2q$, $r' = r/(r-1)$. Choose α so that $\alpha r' > d$. The environment–site intertwining and heat-kernel symmetry give, for each coordinate derivative,

$$\left| D e^{-s\mathcal{L}_L} F \right| \leq \left(\sum_y [\omega_{L,s}(y)^\alpha |\nabla_y G_L(s, J, y, 0)|]^r \right)^{1/r} \left(\sum_y \omega_{L,s}(y)^{-\alpha r'} |F_y|^{r'} \right)^{1/r'}. \quad (122)$$

By theorem A.2, the first factor is bounded by

$$C(1+s)^{-d/2-1/2+d/(2r)} e^{-cs/L^2}. \quad (123)$$

For $2p \geq r'$, Minkowski and stationarity yield

$$\left\| \left(\sum_y \omega_{L,s}(y)^{-\alpha r'} |F_y|^{r'} \right)^{1/r'} \right\|_{L^{2p}(\Omega_L)} \quad (124)$$

$$\leq \left(\sum_y \omega_{L,s}(y)^{-\alpha r'} \right)^{1/r'} \|F\|_{L^{2p}(\Omega_L)} \quad (125)$$

$$\leq C \min\{(1+s)^{d/(2r')}, L^{d/r'}\} \|F\|_{L^{2p}(\Omega_L)}. \quad (126)$$

The final sum estimate follows from $\alpha r' > d$ by comparison with the corresponding radial integral on the diffusive ball, capped by the torus volume. Since $1/r + 1/r' = 1$, the powers cancel exactly:

$$-\frac{d}{2} - \frac{1}{2} + \frac{d}{2r} + \frac{d}{2r'} = -\frac{1}{2}.$$

For $s \leq L^2$ use the diffusive-volume term in (126); for $s \geq L^2$ use $L^{d/r'} \leq (1+s)^{d/(2r')}$, while the exponential only improves the bound. Consequently

$$\left\| D e^{-s\mathcal{L}_L} F \right\|_{L^{2p}(\Omega_L)} \leq C_p (1+s)^{-1/2} \|F\|_{L^{2p}(\Omega_L)}. \quad (127)$$

Apply this with $F = u(t/2) = e^{-(t/2)\mathcal{L}_L} d_A$ and $s = t/2$. Equation (120) then gives

$$\left\| D e^{-t\mathcal{L}_L} d_A \right\|_{L^{2p}} \leq C_p |A| (1+t)^{-1-d/4}. \quad (128)$$

Under the exact stationary extension, this derivative is the physical oriented edge gradient $w_{x,\mu}(t) = b_{x,\mu}^* e^{-tL_{J,L}} g_L$. Spatial stationarity and restoration of j_+ prove the periodic gradient-propagator estimate in the main text for $p = 1, 2$.

B Reproducibility manifest

The Paper II directory contains the TeX and PDF, the seven fluctuation and sharp-cutoff verifiers, retained CSV/PNG/JSON certificates, the detailed gradient-propagator source map, a claim-status file, and SHA-256 checksums. The root archive contains Paper I and the reviewer-response matrix.

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